

MULTI-SENSOR GAIT DATA COLLECTION & PREPROCESSING PIPELINE

8/25/2025

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Project Overview

Final Goal of the Project:

- Develop a neural network model to predict joint trajectories for one future gait cycle

Objectives:

- Collect gait cycle data using the Vicon motion capture system, EMG sensors, and IMU sensors
- Preprocess the collected data through gap filling, labelling, filtering, and syncing
- Train a neural network model to generate joint trajectory predictions



Equipment & Devices

1. Vicon Motion Capture System
 - Vicon Nexus Software
 - Cameras
 - Markers
2. EMG Sensors
3. IMU Sensors
4. Treadmill
5. Bertec Force Plates



Figure 1: Neuromechanics and Assistive Robotics Lab

DATA COLLECTION

Data Collection

URA Tasks:

- Gathered participants
- Assisted with the markers and sensors attachment
- Measured participant demographics
- Vicon camera setup
- Assisted static & functional calibration
- Monitored trials to identify any issues arising during the data collection

Data Collection – Sensors Attachment

IMU:

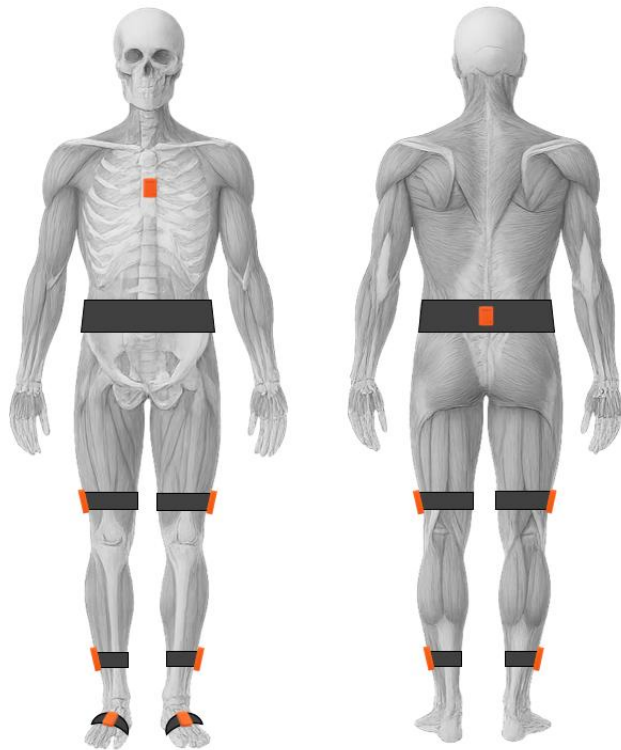


Figure 2: IMU Sensors Attachment Guideline

EMG:

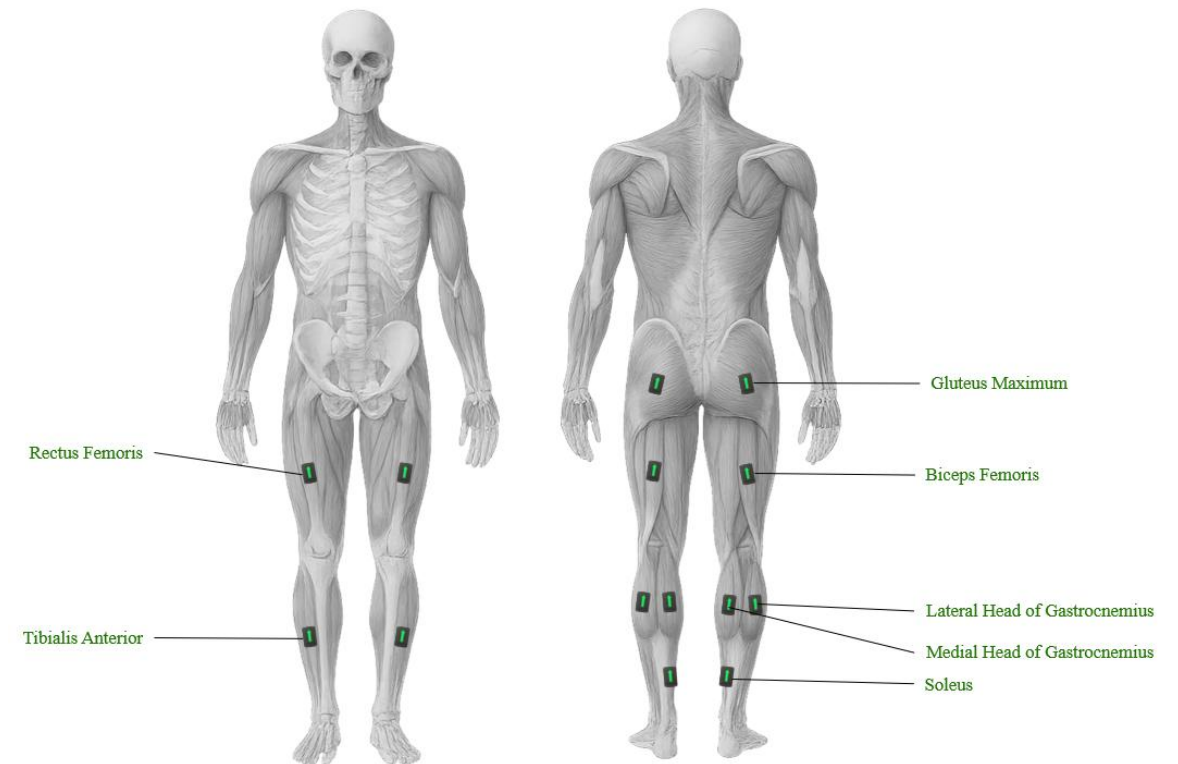


Figure 3: EMG Sensors Attachment Guideline

PREPROCESSING STEPS

Gap Filling, Labelling, model outputs, EMG disconnections, IMU syncing

Gap Filling

Table 1: Vicon Nexus Gap Filling Options

Method	Description	Use Case
Spline fill	Cubic interpolation between points	Small gaps (<10 frames)
Cyclic fill	Patterns from earlier or later gait cycles	Longer, cyclic data gaps
Rigid body fill	Uses 3 source markers from the same rigid body to estimate the missing one	Rigid body structures (E.g., hip markers)
Pattern fill	Uses the pattern of a nearby reference marker in the same segment to estimate the missing marker	When the inner marker is covered but the outer marker is present, or vice versa (E.g., RKNM & RKNE)
Kinematic fill	Based on the kinematic fit skeleton & segment relationships from the model	Barely used

Labelling & Model Outputs

- Knee medial & Ankle medial markers were added to accurately capture the axes of rotation of the knee and ankle.
- Match the local coordinate system of knee/ankle axis with the medial & lateral rotation axes.

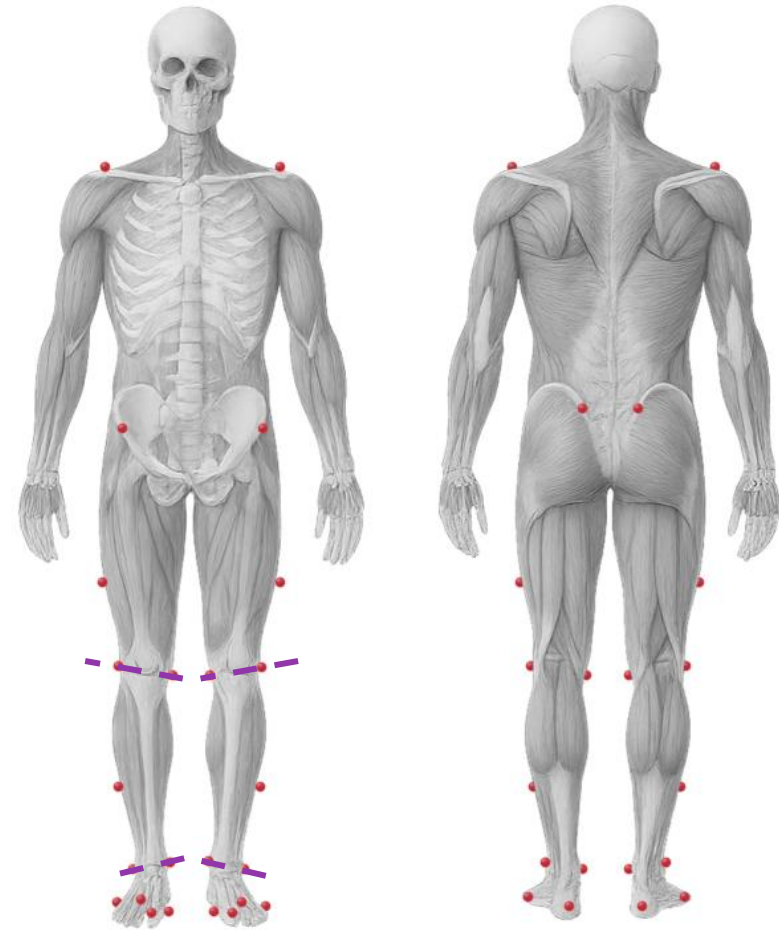


Figure 4: Vicon Marker Attachment Guideline

EMG Data Disconnection

- Issue: EMG sensor disconnection → unusable data segments
- Solution: Replace disconnected regions with linear interpolation between valid signal points before and after the disconnection and apply a special spectrum.

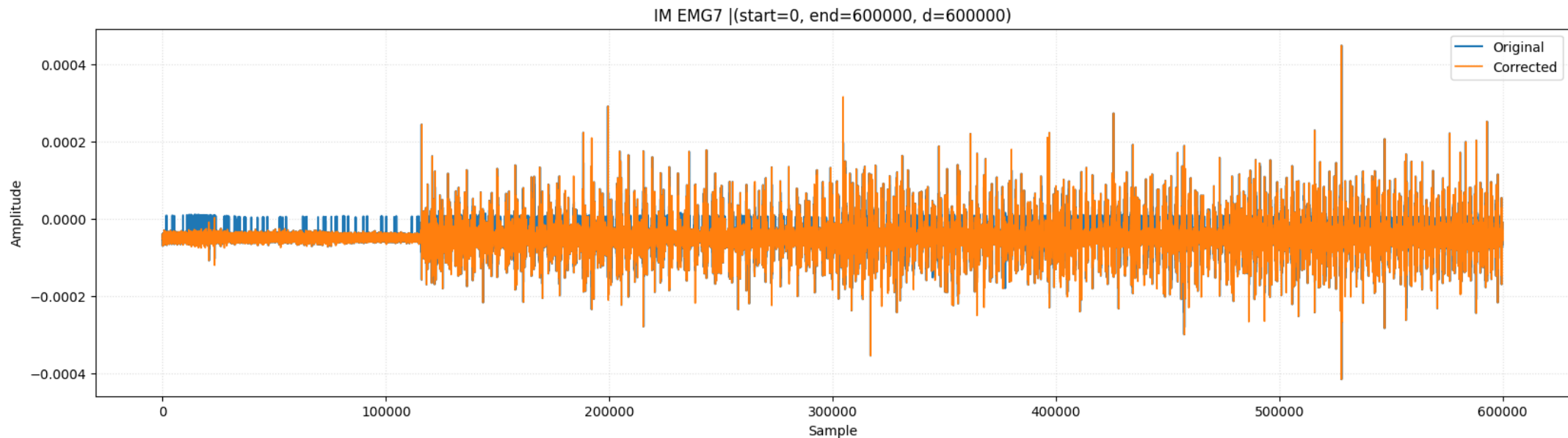
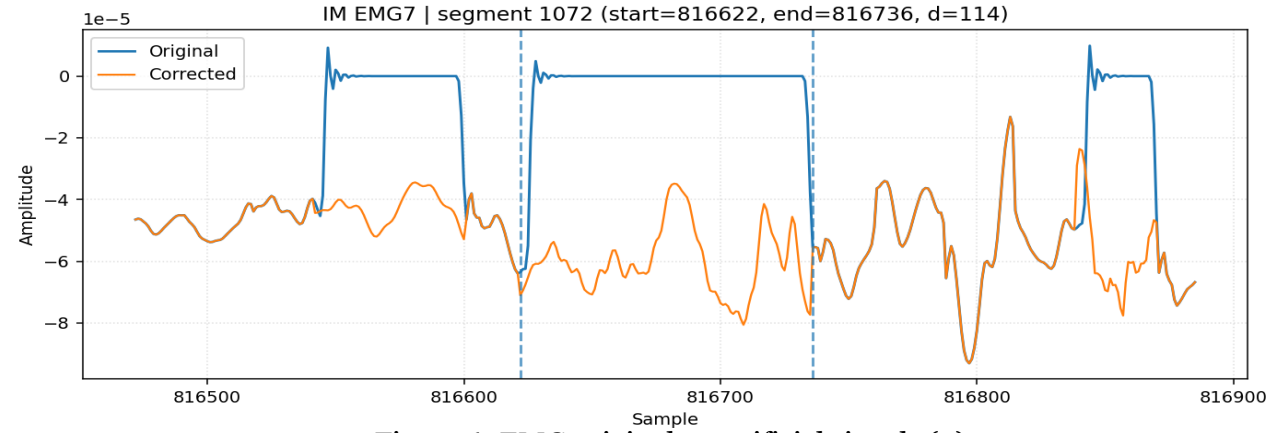


Figure 5: EMG original vs artificial signals (1)

EMG Data Disconnection

1. **Two Thresholds:** Define two thresholds and flag frames that fall within the range.
2. **Slope Detection:** Identify the start and end points of falling/rising slopes
3. **Linear Interpolation:** Connect the start and end points of the disconnection region with a straight line
4. **Standard Deviation:** Calculate the standard deviation of one window before and after disconnection
5. **Fast Fourier Transform (FFT):** Apply FFT to one window before and after to convert the time domain to the frequency domain.
6. **Artificial Random Signal Generation:** Generate target signals using the mean frequency from the regions before and after, standard deviation of 1, and a mean of 0.
7. **Final Generated Signal:** Multiply the standard deviation of the specific point by the randomly generated signal and add it to the linear interpolation line.



IMU Syncing

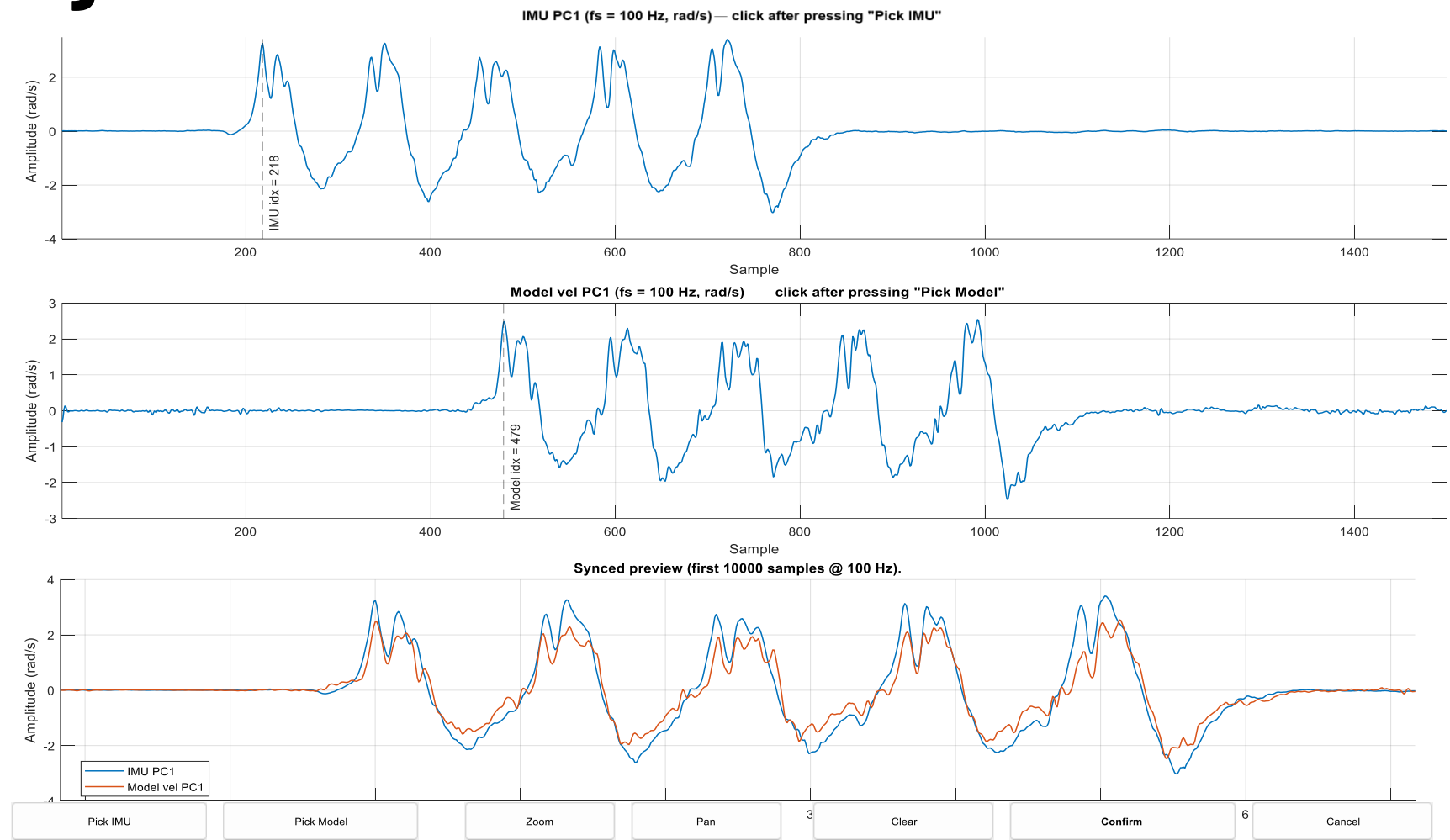


Figure 7, 8, 9: IMU Syncing (1)

IMU Syncing

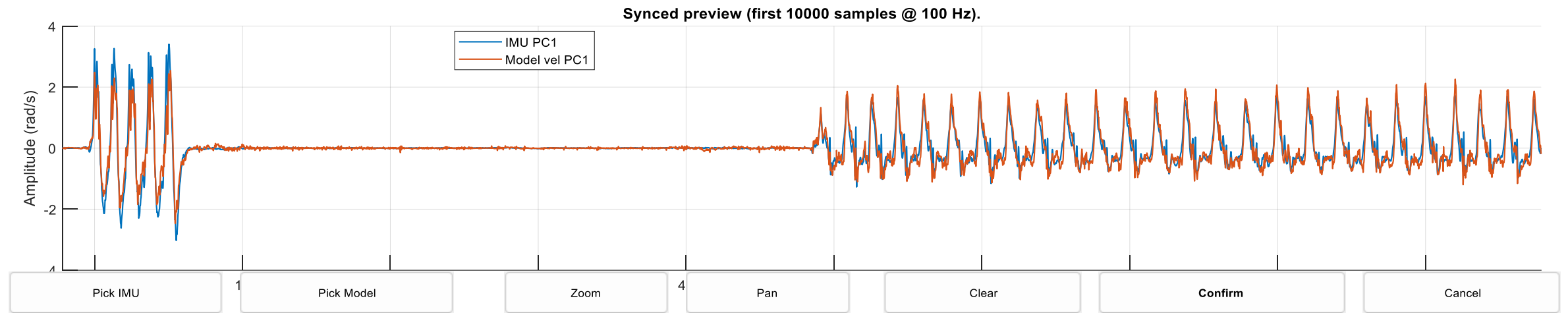


Figure 10: IMU Syncing (2)

CONCLUSION

What I Learned

- Biomechanics concepts: gait cycle, lower limb muscle groups, axes of rotation of joint angles, etc.
- Mathematical concepts: Euler angle, signal filtering backend, Principal Components Analysis (PCA), Fast Fourier Transformation (FFT), etc.
- Research Process: designing the experiments, data collection, preprocessing data, applying mathematical tools to solve issues, and analyzing results

Next Steps

- More preprocessing on the EMGs, force signals, IMU signals data including filtering and segmentation
- Learning about machine learning and neural network concepts
- Train neural network model

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